

Lab 3 – Architectural Justification

SRN: PES1UG24AM221

System: Domain & SSL Certificate Expiry Alert System

Architecture Selection

We chose Microservices Architecture for the Domain & SSL Certificate Expiry Alert System.

Architectural Choice

The Microservices Architecture is the most suitable architecture for the Domain & SSL Certificate Expiry Alert System because the system contains several independent monitoring and processing functions, such as WHOIS auditing, SSL/TLS auditing, expiry detection, and alert management.

Reason 1 – Independent Monitoring Services

WHOIS auditing and SSL/TLS certificate auditing perform different monitoring tasks and can operate as independent services. With a microservices architecture, the **WHOIS Audit Service** and **SSL/TLS Audit Service** can be developed, maintained, and executed independently without affecting the other monitoring service.

Reason 2 – Scalability and Fault Isolation

The number of monitored domains may increase over time. Microservices allow individual services to be scaled according to their workload. For example, the monitoring services can handle a larger number of domain and certificate checks without requiring the entire system to be scaled. If one monitoring service experiences a failure, the other services can continue operating, improving fault isolation.

Security Advantage

Microservices allow security responsibilities to be separated between services. Access to the **Web Application** and administrative functions can be secured using authenticated HTTPS communication, while sensitive domain and audit information can be isolated within the appropriate services and database. This reduces unnecessary access between system components.

Performance Benefit

WHOIS and SSL/TLS audits can be performed independently and potentially in parallel. This reduces the time required to monitor multiple domains and allows the system to process monitoring tasks more efficiently.

Conclusion

Microservices Architecture provides a suitable balance of **modularity, scalability, fault isolation, security, and performance** for the Domain & SSL Certificate Expiry Alert System.